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Version 1.2
1 May 2019



Introduction

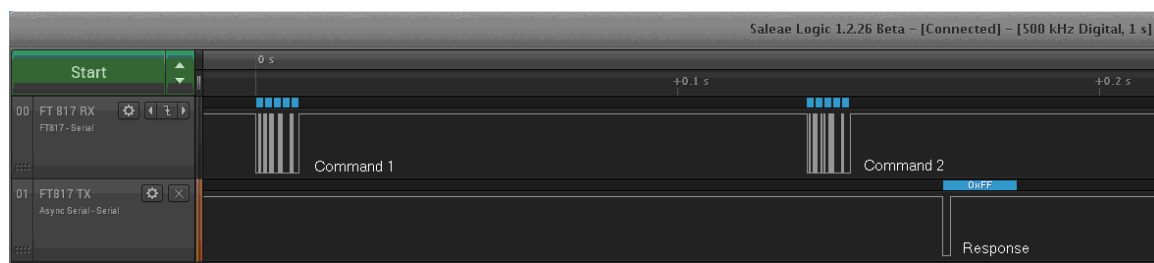
This document describes a simple troubleshooting activity that demonstrates how a cheap logic analyzer can assist in software development.

I used a clone 8 bit logic analyzer to monitor the serial lines from my Arduino to the radio. The software was set up to take 500K samples per second and set to trigger when the first command is detected on the radio receive line.

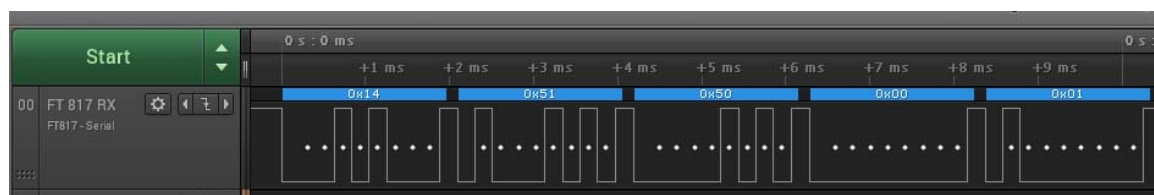
Analyzers like this generally use a CY7C68013A which is an 8051 microprocessor with a USB transceiver. A quick search will return lots of info about these devices. They are cheap and the software is easy to use.

Scenario

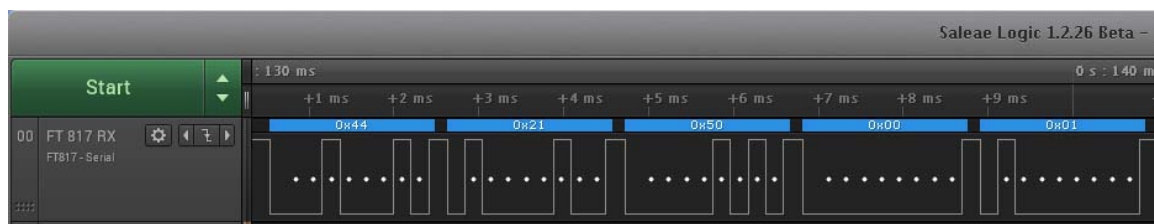
I was writing some code to control a FT817 via a multi-pole rotary switch. When switching from one band to another, the rig occasionally landed on the wrong frequency.



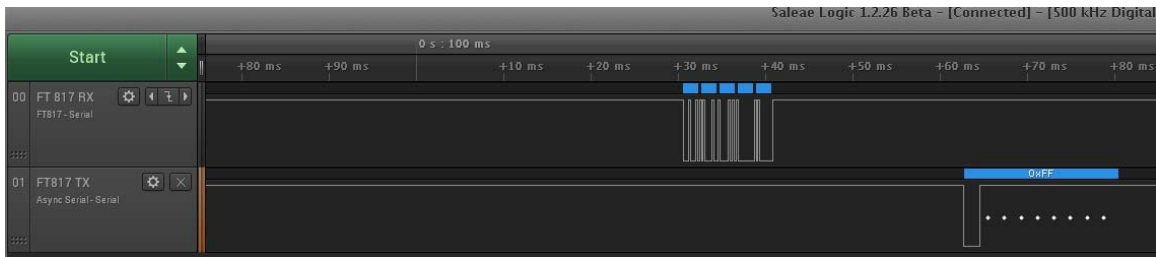
The FT817 was receiving two commands during switching and displays the wrong frequency. We can see two commands but only one ACK from the rig.



Here we can zoom in and see the detail of the first command – Set frequency to 145.150 (not what we want).



Here we see the detail of the second command – Set frequency to 442.150 (what we want) but it is not processed because the rig is still responding to the first command.



Here we can see the rig response (0xFF) comes after the second command at around 160mS. It's unlikely you would ever see this without the use of a logic analyzer.

The first command was coming from the initial state of the switch. I assume as I switched to the next position the switch bounced causing the two different commands to be sent. The rig was still busy responding to the first command when the second one came in and was of course missed.

Solution:

I had to ensure that the messages are never sent faster than the rig can process them. The delay in the switch bounce routine was increased to 200mS which limits the speed at which it can detect a change in switch state. This also helped limit the number of frequency changes if someone spins the band switch really quickly.

A clever programmer would have figured out what was happening and simply extended the delay until the problem went away. However I use this simple case to demonstrate the power of a logic analyzer.

More complex troubleshooting and even reverse engineering of software becomes possible with a device like this.